Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/03/2004
PI Name:	Durian, Douglas J. Ph.D.		
Project Title:	The Melting of Aqueous Foams		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-6396	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	1995 Microgravity Fluid Physics NRA-94-OLMSA-05: Research and Flight Experiment Opportunities
Start Date:	07/21/2000	**End Date:**	05/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	3
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2481		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The unusual elastic character of foams will be quantified macroscopically by measurement of the shear stress as a function of static shear strain, shear strain rate, and time following a step strain; such data will be analyzed in terms of a yield stress, a static shear modulus, and dynamical time scales. Microscopic information about bubble packing and rearrangement dynamics, from which these macroscopic non-Newtonian properties presumably arise, will be obtained non-invasively by novel multiple-light scattering diagnostics such as diffusing-wave spectroscopy (DWS). Quantitative trends with materials parameters, such as average bubble size, and liquid content, will be sought in order to elucidate the fundamental connection between the microscopic structure and dynamics and the macroscopic rheology. To exploit rheological and multiple-light scattering techniques, and ultimately microgravity conditions, in order to quantify and elucidate the unusual elastic character of foams in terms of their underlying microscopic structure and dynamics. Special interest is in determining how this elastic character vanishes, i.e., how the foam melts into a simple viscous liquid as a function of both increasing liquid content and shear strain rate. The utility and fascination of foams are derived largely from the surprising fact that they have a solid-like elastic character in spite of being mostly gas with a few percent volume fraction of liquid, but can nevertheless flow under shear. The physical origin of such unusual rheology in terms of microscopic structure and dynamics is poorly understood and remains a subject of basic scientific interest to physicists, chemists, and chemical engineers. The proposed research promises important new insight into these issues, and could also have significant consequences for our understanding of flow in other dense randomly-packed systems such as emulsions, colloidal suspensions, slurries, bubbly liquids, and granular materials. Furthermore, all foam applications are empirically based and the proposed research may generate valuable fundamental guidance for the development of materials with more desirable rheology and better stability.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Foams are important for a clean environment. Firstly, there presence in natural waters signals the presence of surface-active impurities and a need for clean-up. Secondly, they are useful themselves in the cleanup process since many impurities can be removed by injecting gas and skimming off the resulting foam.
Task Progress:	Since October 2003 we have made progress on a number of topics. The design of the flight experiment by engineering companies contracted by ESA is proceeding rapidly. In particular, the Phase-B Final meeting was held at ESA ESTEC Headquarters in Katwijk, Netherlands, on March 11-12 2004. The original platform was to be an Experimental Container in the Fluid Science Laboratory. However, it now seems likely that this will b switched to the Microgravity Science Glovebox for flight at the end of 2006. Besides a more rapid time to flight, the advantages are more space and ready availability of Nitrogen. It now appears that means of foam production by turbulent mixing are now finalized to our satisfaction. The chemical composition of the foaming liquid is also finalized to our satisfaction. The measuring geometry and dimensions are also finalized. One outstanding issue is still foam containment and the implications of friction in seals for doing rheology. Another outstanding issue is foam destruction and recycling of the foaming solution. Postdoc Dr. Pam Korda is now experimenting with a means of doing rheology in a truly closed cell. The idea would be to seal the edge of the rotating cone bob with a flexible rubber sheet. One would then treat the system as a damped torsional oscillator, whose frequency and decay rate depend on the rheological characteristics of the foam. In the lab, Dr. Korda now has a mock system for doing just this. Dr. Korda is also making progress on a means of doing tomography very rapidly on a dry large-celled foams. The idea is to capture not only foam structure, as in the past, but to do so with sufficient speed as to capture the dynamics of bubble rearrangements. These are crucial for understanding how foams melt as a function of strain rate. Graduate student Alex Gittings has now finished a program of measuring higher-order intensity correlations for light diffusely transmitted and backscattered from foams. In the usual illumination/detection geometries to be employed in the flight experiment, we find that the signals are Gaussian. Previously this was assumed, but was never explicitly demonstrated. Furthermore, for a point-in / point-out backscattering geometry we find that the signal is non-Gaussian. This indicates that the statistics of small numbers of rearrangements of bubbles in the scattering geometry is playing a role. In flight experiments, this is to be avoided; now we know how. On earth, one can imagine using this to extract extra information about rearrangements dynamics that will complement/supplement the tomography work of Dr. Korda. In addition, Mr. Gittings is wrapping up a series of computer simulations of photon transport in model foam structures. The issue under investigation is the relative role of films vs borders vs vertices in scattering the light, as well as the importance of borders in channeling the light. By including films, Mr. Gittings is finding that films play a subtle non-negligle role that may explain the curious scaling of the transport mean free path with bubble size and liquid content. A detailed understanding of photon transport is crucial for analysis of flgiht experiment data.
Bibliography Type:	Description: (Last Updated: 06/12/2025)
Articles in Peer-reviewed Journals	Gopal, A.D.; Durian, D.J. "Relaxing in Foam" Physical Review Letters 91, 188303/1-4 , Oct-2003
Articles in Peer-reviewed Journals	Gittings, A.S.; Bandyopadhyay, R.; Durian, D.J. "Photon chanelling in foams" Europhysics Letters 65, 414-9 , Feb-2004
Presentation	Durian, D.J. "Physics of Foams" Sigma-Xi public lecture at Corning Research Corporation Apr-2004